Edward Silva

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Experience

Embedded Software Engineer on the Phased Array Software Team, SpaceX / Starlink — Jul 2026 – Present
C, C++, Python, Git — Bastrop, TX

Software Engineering Intern, Kratos Defense — Jun – Aug 2025
C++, Python, SIMD, Git — Colorado Springs, CO

- Sped up legacy real-time DSP code in C++ by 1.6x through refactoring, AVX-512 vectorization, and Intel VTune profiling on Linux over SSH
- Presented a faster FIR filter implementation and its performance trade-offs to the team for use on later projects
- Built a configurable logging framework with severity levels and error tracking to cut debug time and improve maintainability of the real-time DSP software

Co-op Intern, Electrical Design, Jordan and Skala Engineers — Jan – Jun 2025
Autodesk Revit, AutoCAD MEP, NEC Compliance — Denver, CO

- Performed circuit loading and voltage drop calculations, sized breakers to NEC requirements, and drafted electrical layouts and design documentation for 20+ multi-unit residential and non-residential projects

Undergraduate Researcher, ePower Hubs Research Lab — Aug – Dec 2024
Golden, CO

- Reviewed the literature on sensor systems and farm-level control strategies for offshore wind, under variable voltage, power, and frequency constraints
- Wrote the findings up in internal LaTeX reports on cost reduction in wind farm maintenance, design, and grid integration

Undergraduate Researcher, Explosives Research Lab — Fall 2022
Golden, CO

Projects

Electromagnetic Bicycle Damper (EMBR), MATLAB, Python — Fall 2025 – Spring 2026

- Owned the electromagnet design and calculation verification within an 8-person interdisciplinary team, sponsored by ARUS MR Tech, that retrofitted a production front suspension fork with a semi-active MR damper
- Modeled magnetic field strength and spatial geometry using the Biot-Savart law and complete elliptic integrals in Python, ensuring field lines oriented perpendicularly to fluid flow to maximize yield stress activation

Non-Linear State Estimation for the EMBR Damper, MATLAB, Simulink — Spring 2026

- Derived continuous and discrete-time non-linear state-space models in MATLAB and Simulink, simulating system kinematics to validate damping performance against harmonic, stochastic, and discrete terrain disturbances
- Implemented and compared Extended (EKF) and Unscented (UKF) Kalman Filters to estimate unmeasured states

Pitch-Tracking Systems in Auto-Tune, MATLAB — Spring 2026

- Measured latency and residual error to verify the performance of real-time Fast Fourier Transform (FFT) and Autocorrelation (ACF) pitch-tracking algorithms in MATLAB
- Designed automated data visualization scripts to quantify computational processing time, demonstrating a 2.68x execution speedup for FFT over ACF

Gantry Crane Control System, MATLAB, Simulink — Fall 2025

- Designed a 50 Hz discrete-time controller for a gantry-crane system by linearizing the plant model, constraining cart velocity to <= 1 m/s and achieving 0% overshoot for 1 m position step commands
- Verified system robustness through frequency-domain analysis, proving stability margins including a 48° phase margin and a ~0.5 s delay tolerance

Autonomous Maze-Navigating Robot, Python Fall 2025

- Achieved the second-fastest course completion time in the cohort with an autonomous maze-navigating robot built by a 4-person team, integrating computer vision, embedded control, and I2C communication
- Built the computer vision subsystem with one teammate, using Python and OpenCV with ArUco marker detection to extract real-time distance and angle metrics for navigation

Hydraulic Ram Pump Fall 2022

- Built a working hydraulic ram pump prototype for under $100 within a 5-person team, using PVC piping and a pH filtration system to mitigate acid mine drainage at legacy mine sites
- Presented the design to a panel of industry professionals, competing against dozens of teams
- Received the Subject Matter Expert Award at the final presentation

eeprep, GitHub Ongoing

- Wrote eeprep, a free guide for EECS students and new graduates looking for a first job

resume-generator, GitHub Ongoing

- Built a script that turns one JSON file into a one-page PDF resume, fits the page from LuaLaTeX measurements, and stops when the page will not fit

company-lists, GitHub Ongoing

- Publish an open dataset of companies by region, starting with Denver, Colorado, with a public source for every row

Skills

Skills: C++, Python, C, SIMD, MATLAB, Simulink, Git, Arduino

Education

Colorado School of Mines May 2026

BS Electrical Engineering, Minor in Computer Science, Golden, CO

Honors: Dean's List (Fall 2022, Fall 2024, Spring 2026) | Honor Roll (Fall 2023, Spring 2024, Fall 2025)

Scholarships: Provost Scholarship (Merit scholarship) | 2023 - 2024 C-MAPP Scholar (Frontier Technology Inc.) | 2024 - 2025 C-MAPP Scholar (Parsons Corporation) | 2024 - 2025 American Bureau of Shipping Scholar (Merit scholarship)

Coursework: CSCI 406 Algorithms, EENG 411 Digital Signal Processing, EENG 519 Estimation Theory & Kalman Filtering, EENG 417 Modern Control Design, EENG 383 Embedded Systems, CSCI 341 Computer Organization, CSCI 306 Software Engineering, EENG 393 Field Session – Printed Circuit Board Design, EENG 350 Electrical Engineering Integration and Design, CSCI 220 Data Structures and Algorithms, EENG 307 Introduction to Feedback Control Systems, EENG 391 Field Session – Computational Methods for Electrical Engineers, EENG 385 Electronic Devices and Circuits, EENG 386 Fundamentals of Engineering Electromagnetics, EENG 282 Electrical Circuits Laboratory, EENG 284 Digital Logic, CSCI 410 Elements of Computing Systems, CSCI 358 Discrete Mathematics, CSCI 200 Foundational Programming Concepts & Design, CSCI 274 Introduction to the Linux Operating System, EENG 310 Information Systems Science I, EENG 311 Information Systems Science II, EENG 389 Fundamentals of Electric Machinery, CSCI 290 Programming Challenges I

Certifications

MATLAB Machine Learning Techniques, MathWorks
MTA: Introduction to Programming Using Java, Microsoft
MTA: Introduction to Programming Using Python, Microsoft

Organizations

IEEE-HKN, Nu Pi Chapter, Colorado School of Mines, Member 2026 – Present
IEEE, Member 2025 – Present
IEEE Young Professionals, Member 2025 – Present
Denver Astronomical Society, Member 2022 – 2025